

Muhammad Shaheer Saikhani

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EDUCATION

FAST – NUCES, Lahore

2024 – Present

Bachelor of Science in Computer Science | Currently in 4th Semester | GPA: 3.38 / 4.00

Relevant Courses: Programming Fundamentals, OOP, Data Structures & Algorithms, Database Systems, Computer Organization & Assembly Language, Software Design & Analysis

Punjab Group of Colleges, Lahore

Completed

Intermediate – ICS (Intermediate in Computer Science)

The Punjab School, Lahore

Completed

Matriculation (SSC)

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, JavaScript, Assembly Language (MASM)

Frameworks & Technologies: Node.js, Express, React, FastAPI, MSSQL, Docker

AI / ML: XGBoost, scikit-learn, SHAP, OpenAI API, Agentic AI

Concepts: OOP, Data Structures, Database Design, Software Design Principles (DIP, Observer Pattern)

Tools: Git, GitHub, VS Code, Visual Studio

PROJECTS

MaternaSafe – AI Maternal Health Copilot

National AI Hackathon 2026 | Python, FastAPI, React, XGBoost

- Built an AI-powered maternal health risk assessment system during a competitive two-day national hackathon attended by teams from across Pakistan.
- Designed a rule-based + ML risk engine (XGBoost) that assesses vitals (blood pressure, glucose, heart rate, BMI) and produces a risk score with SHAP-style explanations.
- Developed a FastAPI microservices architecture with separate ML and backend services, integrated with OpenAI API to generate personalised recommendations and Urdu-language summaries.
- Implemented a React + Tailwind frontend with multi-page assessment flow, chat interface, and voice input support.

GDP Analysis & Real-Time Data Pipeline

Software Design & Analysis Project | Three-Phase

- Phase 1: Dashboard that reads GDP data from CSV and displays key statistics and trends through visual charts.
- Phase 2: Refactored codebase applying the Dependency Inversion Principle (DIP) to separate system components, improving maintainability.
- Phase 3: Designed a real-time data pipeline with three sequential modules — raw input, authentication via security hash matching, and running average computation with live dashboard output. Implemented Observer Pattern to monitor packet queue sizes.

Online Library Management System

Database Systems & Software Design | Node.js, React, MSSQL

- Built a full-stack library management website with a Node.js/Express backend, MSSQL database, and React frontend.
- Enables users to browse, manage, and track books through a clean and responsive web interface.

Flight Reservation System

Data Structures & Algorithms | C++

- Console-based flight reservation app using core data structures; supports adding flights, reserving/cancelling seats, and managing bookings through a text-based menu.

Popping Balloon Game

Computer Organization & Assembly Language | x86 Assembly

- Interactive typing game in Assembly; letters appear inside balloons and pressing the correct key pops them with a sound effect. Includes score tracker, 5-life system, and separate game screens.

Tetris & Ping Pong Games

OOP & Programming Fundamentals | C++

- Tetris: fully playable using OOP (classes, inheritance, encapsulation) with falling blocks, score display, and next-block preview.
- Ping Pong: single-player game with wall collision detection, live score tracking, and console-based real-time rendering.

ACHIEVEMENTS & CERTIFICATIONS

Dean's List of Honors

Fall 2025 | FAST NUCES Lahore

Certificate of Participation – National AI Hackathon 2026

6–7 May 2026 | Lahore

Awarded by atomcamp, PSDF & NUCES Fintech Society for participating in the Two-Day Agentic AI Workshop held at the National AI Hackathon 2026, Lahore — a competitive national-level event.

COMPETITIVE PROGRAMMING

LeetCode – Solved 32 problems in C++ (Easy: 23, Medium: 9) | leetcode.com/u/shaheer-saikhani

GitHub – Active repositories including collaborative team projects | github.com/Shahreer-Saikhani

INTERESTS

Problem Solving, Football, Table Tennis